End of Year Maths Assessment

Year 6

This resource corresponds with the New National Curriculum objectives for Year 6, and has been designed to be used as an aide in assessing pupils' baseline mathematical knowledge at the end of Year 6.

This baseline assessment can be used to assist teachers with end of year ability levelling, planning/assessment and as an ideal support tool for parents' evenings/progress meetings etc.

The content addresses each curricular objective outlined in the New National Curriculum in chronological order to give insight into strengths or gaps in each child's mathematical knowledge.

More Assessment resources.

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End of Year Maths Assessment – Year 6

Year 6 National Curriculum Objectives:

Number – number and place value

1, 2, 3 & 4. (6N2) Read, write, order and compare numbers up to 10 000 000

8. (6N3) Determine the value of each digit in numbers up to 10 000 000

5. (6N4) Round any whole number to a required degree of accuracy

6 & 7. (6N5) Use negative numbers in context, and calculate intervals across zero

1, 2, 3, 4, 5, 6, 7 & 8. (6N6) Solve number and practical problems that involve 6N2 - 6N5

Number – addition, subtraction, multiplication and division

20. (6C3) Use estimation to check answers to calculations and determine, in the context of a problem, an appropriate degree of accuracy

18 & 19. (6C4) Solve addition and subtraction multi-step problems in contexts, deciding which operations and methods to use and why

14, 15 & 16. (6C5) Identify common factors, common multiples and prime numbers

13. (6C6) Perform mental calculations, including with mixed operations and large numbers

9 & 10. (6C7a) Multiply multi-digit numbers up to 4 digits by a two-digit whole number using the formal written method of long multiplication

12. (6C7b) Divide numbers up to 4 digits by a two-digit whole number using the formal written method of long division, and interpret remainders as whole number remainders, fractions, or by rounding, as appropriate for the context

11. (6C7c) Divide numbers up to 4 digits by a two-digit number using the formal written method of short division where appropriate, interpreting remainders according to the context

17, 18 & 19. (6C8) Solve problems involving addition, subtraction, multiplication and division

17. (6C9) Use their knowledge of the order of operations to carry out calculations involving the four operations

End of Year Maths Assessment – Year 6

Year 6 National Curriculum Objectives:

Number – fractions (including decimals and percentages)

21. (6F2) Use common factors to simplify fractions; use common multiples to express fractions in the same denomination

22 & 23. (6F3) Compare and order fractions, including fractions > 1

24. (6F4) Add and subtract fractions with different denominators and mixed numbers, using the concept of equivalent fractions

25. (6F5a) Multiply simple pairs of proper fractions, writing the answer in its simplest form [for example, 1/4 × 1/2 = 1/8]

26. (6F5b) Divide proper fractions by whole numbers [for example, 1/3 ÷ 2 = 1/6]

27. (6F6) Associate a fraction with division and calculate decimal fraction equivalents [for example, 0.375] for a simple fraction [for example, 3/8]

28. (6F9a) Identify the value of each digit in numbers given to three decimal places and multiply and divide numbers by 10, 100 and 1,000 giving answers up to three decimal places

29. (6F9b) Multiply one-digit numbers with up to two-decimal places by whole numbers

30 & 31. (6F9c) Use written division methods in cases where the answer has up to two decimal places

32 & 33. (6F10) Solve problems which require answers to be rounded to specified degrees of accuracy

34. (6F11) Recall and use equivalences between simple fractions, decimals and percentages, including in different contexts

Ratio and Proportion

35 & 36. (6R1) Solve problems involving the relative sizes of two quantities where missing values can be found by using integer multiplication and division facts

37 &38. (6R2) Solve problems involving the calculation of percentages [for example, of measures, and such as 15% of 360] and the use of percentages for comparison

39. (6R3) Solve problems involving similar shapes where the scale factor is known or can be found

40. (6R4) Solve problems involving unequal sharing and grouping using knowledge of fractions and multiples

End of Year Maths Assessment – Year 6

Year 6 National Curriculum Objectives:

Algebra

41, 44 & 45. (6A1) Express missing number problems algebraically

41 & 42. (6A2) Use simple formulae

42 & 43. (6A3) Generate and describe linear number sequences

46. (6A4) Find pairs of numbers that satisfy an equation with two unknowns

47. (6A5) Enumerate possibilities of combinations of two variables

Measurement

48, 49, 50, 51. (6M5) Use, read, write and convert between standard units, converting measurements of length, mass, volume and time from a smaller unit of measure to a larger unit, and vice versa, using decimal notation to up to three decimal places

52. (6M6) Convert between miles and kilometres

53. (6M7a) Recognise that shapes with the same areas can have different perimeters and vice versa

56 & 57. (6M7b) Calculate the area of parallelograms and triangles

54 & 55. (6M7c) Recognise when it is possible to use formulae for the area of shapes

58. (6M8a) Calculate, estimate and compare volume of cubes and cuboids using standard units, including cubic centimetres (cm3) and cubic metres (m3), and extending to other units [for example, mm3 and km3]

58. (6M8b) Recognise when it is possible to use formulae for the volume of shapes

48, 49, 50, 51. (6M9) Solve problems involving the calculation and conversion of units of measure, using decimal notation up to three decimal places where appropriate

End of Year Maths Assessment – Year 6

Year 6 National Curriculum Objectives:

Geometry – properties of shapes

61. (6G2a) Compare and classify geometric shapes based on their properties and sizes

*** (6G2b)** Describe simple 3-D shapes

59 & 60. (6G3a) Draw 2-D shapes using given dimensions and angles

*** (6G3b)** Recognise and build simple 3-D shapes, including making nets

62. (6G4a) Find unknown angles in any triangles, quadrilaterals, and regular polygons

64. (6G4b) Recognise angles where they meet at a point, are on a straight line, or are vertically opposite, and find missing angles

63. (6G5) Illustrate and name parts of circles, including radius, diameter and circumference and know that the diameter is twice the radius

Geometry – position and direction

66 & 67. (6P2) Draw and translate simple shapes on the coordinate plane, and reflect them in the axes

65 & 68.(6P3) Describe positions on the full coordinate grid (all four quadrants)

Statistics

69 & 70. (6S1) Interpret and construct pie charts and line graphs and use these to solve problems

71. (6S3) Calculate and interpret the mean as an average

End of Year Maths Assessment – Year 6

Number and Place Value

1. Write the following numbers in words.

6,540,100 _____

8,111,093 _____

2. Write the follow numbers in digits.

Three hundred and sixty-five thousand, nine hundred and two. _____

Four million, eight hundred and ten thousand, four hundred and seventy-two. _____

3. Use the symbols > or < to compare these numbers.

5,645,011 ☐ **5,445,011** **1,087,998** ☐ **1,087,990** **987,890** ☐ **987,990**

4. Order the following numbers from smallest to largest.

909,990	990,009	900,999	909,999	900,009

5. Round the following numbers to the nearest 10.

14 _____ **35** _____ **68** _____

Round the following numbers to the nearest 100.

211 _____ **765** _____ **985** _____

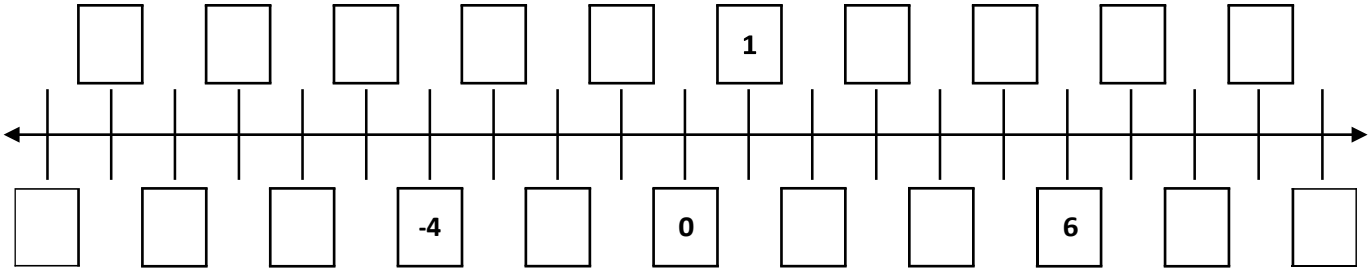
Round the following numbers to the nearest 1,000.

3,987 _____ **2,502** _____ **3,456** _____

Round the following numbers to the nearest 10,000.

15,987 _____ **23,450** _____ **32,099** _____

6. Fill in the missing numbers on the number line.



7. Put these temperatures in order of coolest to warmest.

- 5°C	0°C	- 7°C	5°C	- 15°C

8. Work out the mystery number. The mystery number has 6 digits.

- **The thousands digit is 8.**
- **The hundreds digit is half of the thousands digit.**
- **The ones digit is a quarter of the thousands digit.**
- **The hundred thousands digit is the same as the ones digit.**
- **The tens digit is 3^2.**
- **The ten thousands digit is a prime number bigger than 5 and smaller than 10.**

The mystery number is: ____ ____ ____ ____ ____ ____

Addition, Subtraction, Multiplication and Division

9. 3,467 x 24 =

10. 5,324 x 32 =

11. 4,608 ÷ 18 =

12. 7,056 ÷ 25 =

13. 7 x ☐ = 1,400 ☐ x 300 = 9,000

350 ÷ ☐ = 7 ☐ ÷ 6 = 600

14. Find the common factors of 12 and 30.

15. Find the common multiples of 9 and 12 under 100.

16. List the first 5 prime numbers after 10.

☐ ☐ ☐ ☐ ☐

17. Complete the calculations.

36 + (8 x 9) = _____ 6^2 + 3 + 9 x 2 = _____

(36 + 8) x 9 = _____ 6^2 + (3 + 9) x 2 = _____

18. John weighs 57kg. Alice weighs 16kg less than John. Haroon weighs 5kg more than Alice.

What is the combined weight of John, Alice and Haroon?

19. Hallie has £10 to spend in the shop. She buys 2 pencils which are 75p each, a calculator which is £2.50, she buys 4 books which are £1.10 each and a pack of pens for £0.99. How much change does she get?

20. A plane travels 8,746 miles on Monday, 15,454 miles on Tuesday, 4,587 miles on Wednesday, 7,123 miles on Thursday and 896 miles on Friday. Estimate how many miles the plane flew over the 5 days.

Fractions (including decimals and percentages)

21. Simplify the fractions into the simplest form.

$\frac{12}{36} = ___$ $\frac{10}{90} = ___$ $\frac{3}{12} = ___$ $\frac{9}{45} = ___$ $\frac{14}{56} = ___$

22. Order these fractions from smallest to largest.

$\frac{7}{8}$ $\frac{2}{3}$ $\frac{3}{4}$ $\frac{2}{5}$ $\frac{1}{2}$

☐ ☐ ☐ ☐ ☐

23. Compare these fractions using the correct symbol. <, > or =

$\frac{2}{3}$ ☐ $\frac{7}{12}$ $\frac{2}{5}$ ☐ $\frac{4}{10}$ $\frac{2}{3}$ ☐ $\frac{7}{9}$

24. Complete the following calculations.

$\frac{1}{2} + \frac{3}{8} =$ ☐ $\frac{6}{21} + \frac{3}{7} =$ ☐

$1\frac{2}{5} - \frac{11}{15} =$ ☐ $1\frac{3}{7} - \frac{1}{2} =$ ☐

25. Multiply the following fractions. Write the answers in the simplest form.

$\frac{2}{3} \times \frac{1}{3} =$ ☐ $\frac{1}{4} \times \frac{2}{5} =$ ☐ $\frac{3}{7} \times \frac{1}{3} =$ ☐

26. Divide the following fractions.

$\frac{3}{4} \div 4 = \square$ $\frac{2}{5} \div 3 = \square$ $\frac{3}{8} \div 5 = \square$

27. Draw lines to match the fraction to the equivalent decimal.

0.1	$\frac{3}{10}$
0.3	$\frac{1}{4}$
0.25	$\frac{4}{5}$
0.6	$\frac{6}{10}$
0.8	$\frac{1}{10}$

28. Complete the table below.

	x 10	x 100	x 1,000
1.554			
0.8			
	÷ 10	÷ 100	÷ 1,000
65			
137			

29. Circle the incorrect calculation.

1.36 x 4 = 5.44 2.32 x 6 = 13.85 1.74 x 5 = 8.7

30. 34.8 ÷ 2 =

31. 597.3 ÷ 3 =

32. Carla has collected £553 in sponsorship doing a charity bike ride. She wants to share the money equally between 4 different charities.

How much will each charity get?

33. Emmie runs 7 laps of the football pitch which is 950 metres. How many metres is 1 lap to the nearest metre?

34. Charlie shares his cake with his friends. He gives 25% to William, 0.2 of the cake to Suzie and $\frac{1}{10}$ of the cake to Monty. Charlie gets the rest of the cake. What percentage does Charlie get?

Ratio and Proportion

Use the information below to answer questions 35 and 36.

Greengrass Greengrocers
Carrots - £1.40 per kg
Apples – 4 for £1
Potatoes - £2.25 for 1.5kg bag
Cheese – 60p per 100g

35. Johan buys 2kg of carrots, 2 apples and 500g of potatoes. How much does he pay altogether?

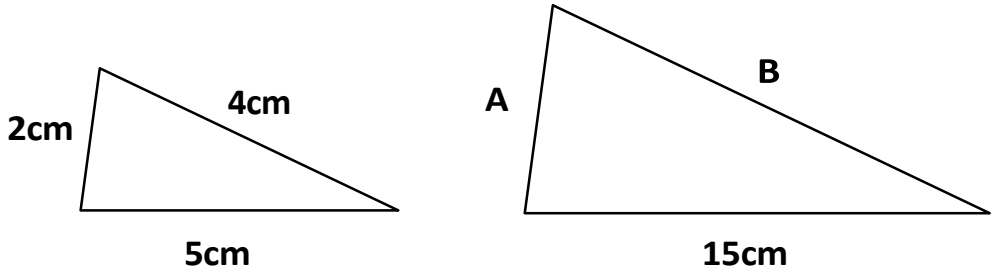
36. Alicia needs 250g of cheese. How much will it cost?

37. Calculate the following percentages.

15% of 200 =	_____	**5% of 80 =**	_____
30% of 75 =	_____	**60% of 350 =**	_____

38. Joe buys a new pair of trainers for £52. He has a 15% off voucher. What is the price of the trainers after he uses the voucher?

39. These shapes are similar. Find the missing lengths.



A = _____ B = _____

What is the scale factor?_____

40. Taku and Arianna have a car-boot sale. They make £240 in total. They share the money between them both but Taku gets £20 more than Arianna. How much do they both get?

Taku gets_____and Arianna gets _____.

Algebra

41. Work out the value of each of the letters.

$a + 17 = 42$	$2n - 16 = 32$	$5d = 50$
$a =$ _____	$n =$ _____	$d =$ _____

42. Create a sequence by completing the table.

n	1	2	3	4	5	6	7	8	9
2n	2	4							
3n + 1	4	7							

43. Work out the missing numbers in each sequence and write the rule.

5 11 [ ] 23 29 [ ] 41 [ ] 53 [ ]

a. The rule is _____

[ ] 50 35 [ ] 5 [ ] -25 [ ] -55 [ ]

b. The rule is _____

2 [ ] 5 9 17 [ ] 65 [ ]

c. The rule is _____

44. Tim is thinking of a number. He multiplies the number by 3 and then adds 21. He gets an answer of 45. What is the number he is thinking of?

45. Ashleigh is thinking of a number. She divides the number by 4 and then adds 11. She gets an answer of 22. What is the number she is thinking of?

46. Find three different solutions to this equation. Numbers must be whole numbers.

$$3a + 2b = 24$$

a = and b =	a = and b =	a = and b =

47. Two numbers when added together make 37. Both numbers are less than 20. What are the two numbers?

and

Measurement

48. Shania is filling up a paddling pool. The pool holds 7.2 litres of water. She is using a cup which holds 120ml.

How many cups full of water will it take to fill the paddling pool?

49. Convert the following litres to millilitres and millilitres to litres.

$1\frac{1}{2}$ l = _____ml $3\frac{1}{4}$ l = _____ ml

1,350 ml = _____l 1,765 ml = _____l

50. Convert the following kilograms to grams and grams to kilograms.

$2\frac{3}{4}$ kg = _____g $5\frac{1}{2}$ kg = _____ g

12,420 g = _____kg 7,650 g = _____kg

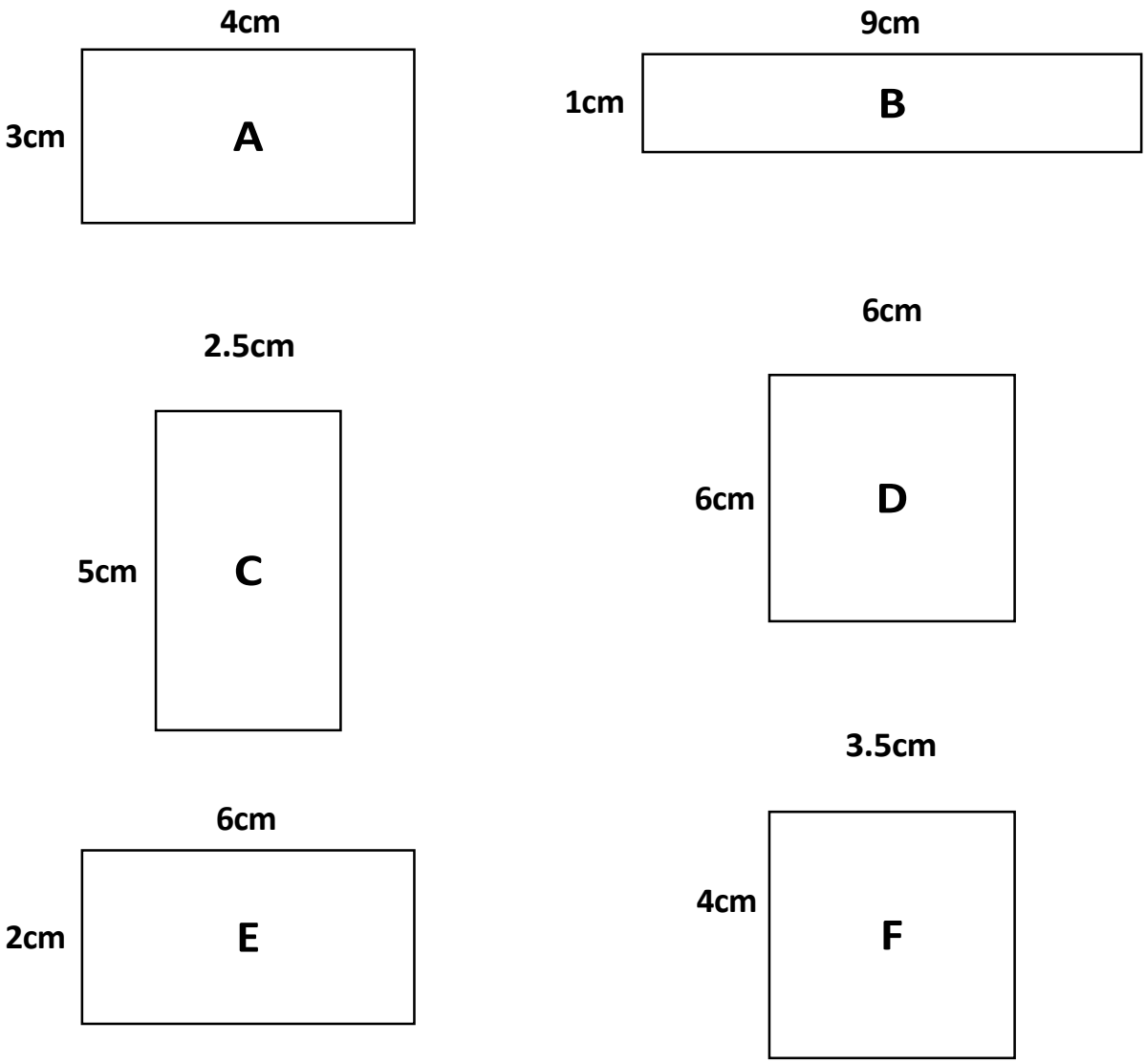
51. Complete the table.

mm	cm	m
	50cm	
		3.5m
250mm		
	310cm	

52. Complete the table.

Approximately 1 mile = 1.6 kilometres	
3 miles	km
miles	8 km
6 miles	km
miles	0.8 km

53. Look at the shapes below.



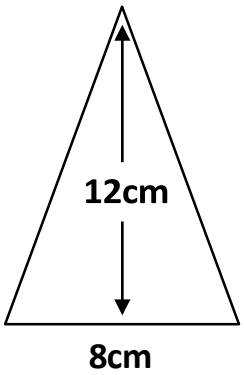
Which two shapes have the same area?

Which two shapes have the same perimeter?

54. The area of a rectangle is 102cm². One side is 6cm, what is the length of the other side? _____

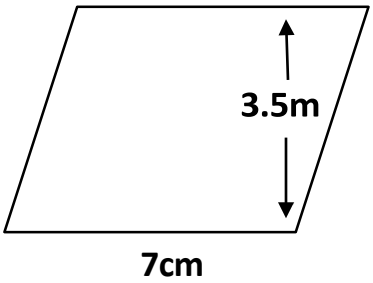
55. The area of a square is 81m². What is the length of one side? _____

56. Calculate the area of the triangle.



Area =

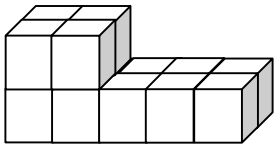
57. Calculate the are of the parallelogram.



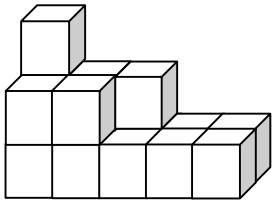
Area =

58. Calculate the volume of the three shapes. Each cube is 1cm³.

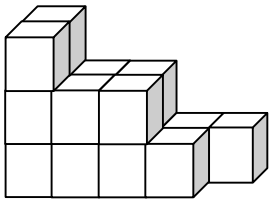
A



B



C



Volume = _____ **Volume = _____** **Volume = _____**

Which shape has the largest volume? _____

Geometry (properties of shapes)

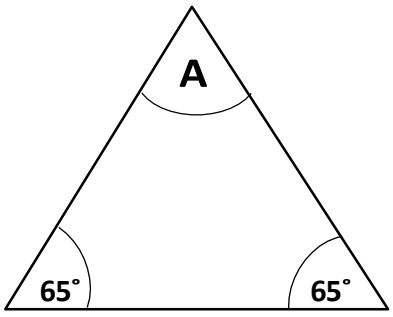
59. Draw a three sided 2D shape which has 2 acute angles, 1 right-angle and 2 sides are equal.

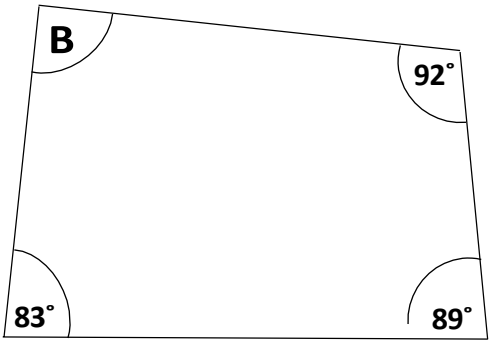
60. Draw a five sided shape which has 2 right-angles.

61. Fill in the properties for the 3D shapes.

3D shape	Faces	Vertices	Edges
Cylinder			
Sphere			
Tetrahedron			
Square Based pyramid			

62. Find the missing angles.

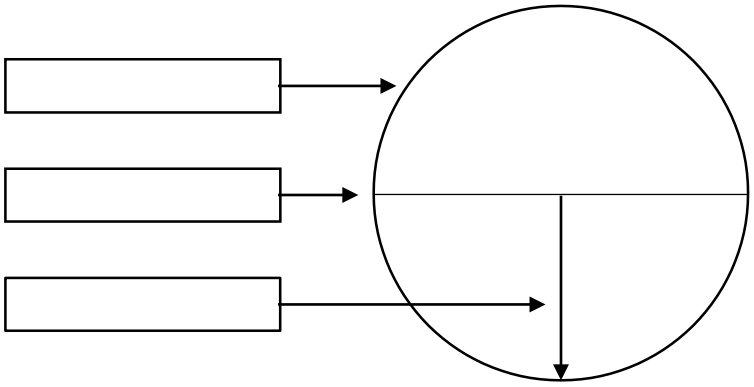




A = _____

B = _____

63. Label the parts of the circle.

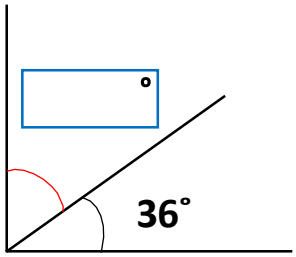


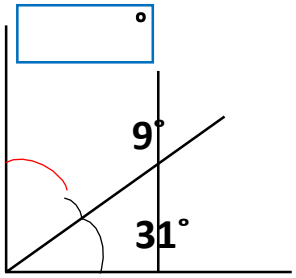
Circumference

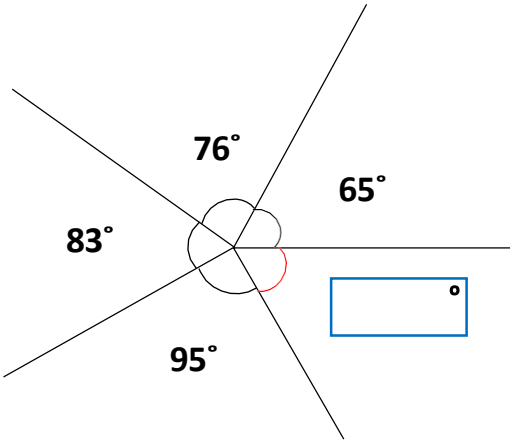
Radius

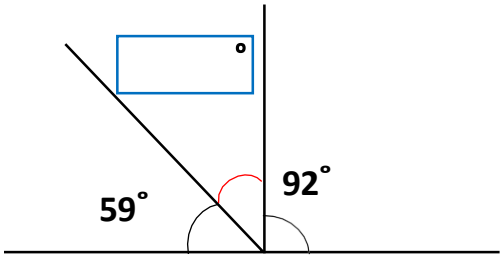
Diameter

64. Calculate the missing angles.



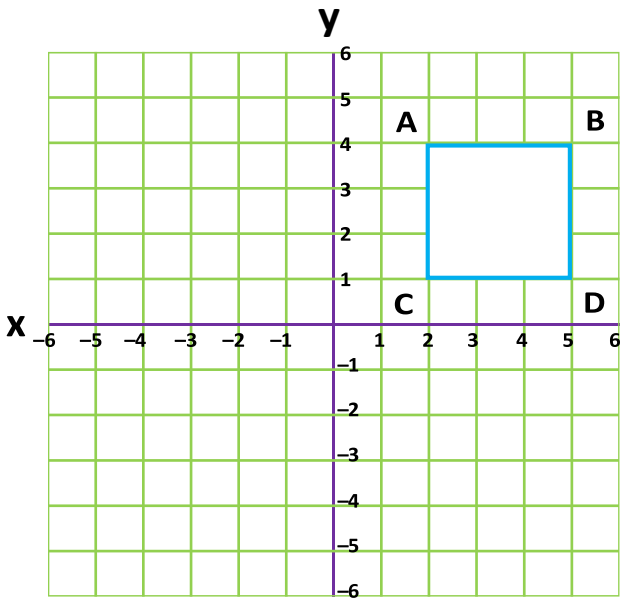






Geometry (position and direction)

65. Write the co-ordinates of the square.



A = _____

B = _____

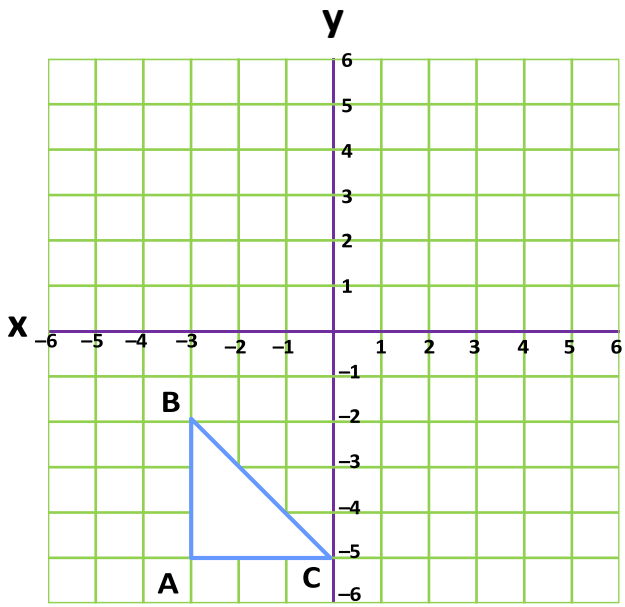
C = _____

D = _____

66. Reflect the square on the y axis and write the new co-ordinates.

A = _____ B = _____ C = _____ D_____

67. Translate the shape.



Instructions

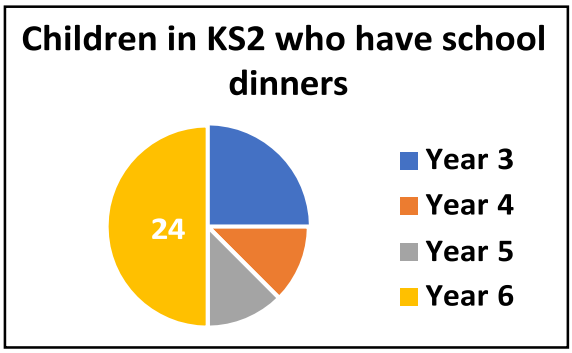
- **Translate the shape 2 squares to the right and 6 squares up.**

68. What are the new co-ordinates

A = _____ B = _____ C = _____

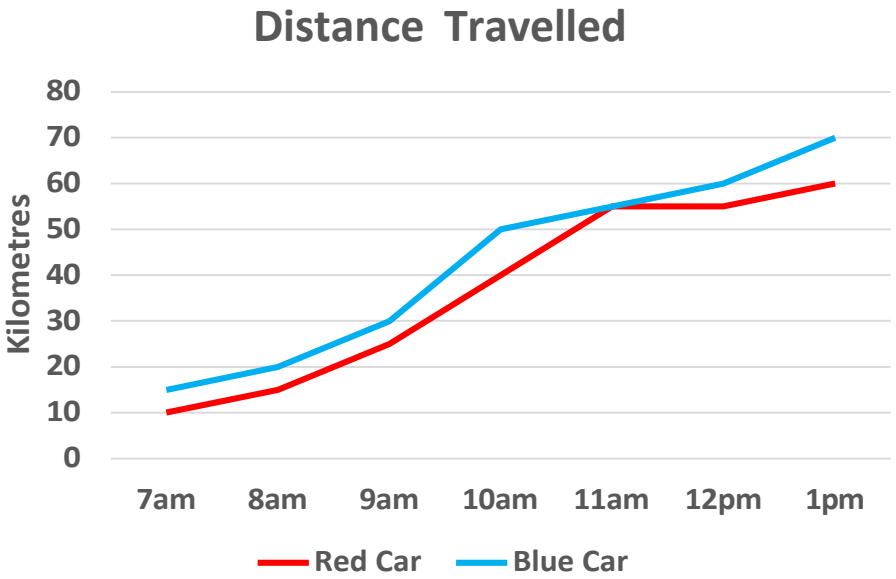
Statistics

69. Use the information from the pie chart.



How many children have school dinners in Year 3? _____
How many children have school dinners in Year 5? _____
How many children in total have school dinners? _____
What fraction of children asked have school dinners in year 4? _____

70.



How far did the red car travel from 7am to 9am? _____

Which car travelled the furthest between 12pm and 1pm?_____

How far did the blue car travel from 10 am to 1pm? _____

How long did it take the blue car to travel 60km from 20km? _____

71. Children in Year 6 are gathering the ages of their parents.

27	37
31	29
31	32
30	29
34	30

What is the mean age? _____

End of Year Maths Assessment – Year 6

Number and Place Value

1. Six million, five hundred and forty thousand and one hundred.
   Eight million, one hundred and eleven thousand and ninety-three.

2. 365; 902

   4,810; 472

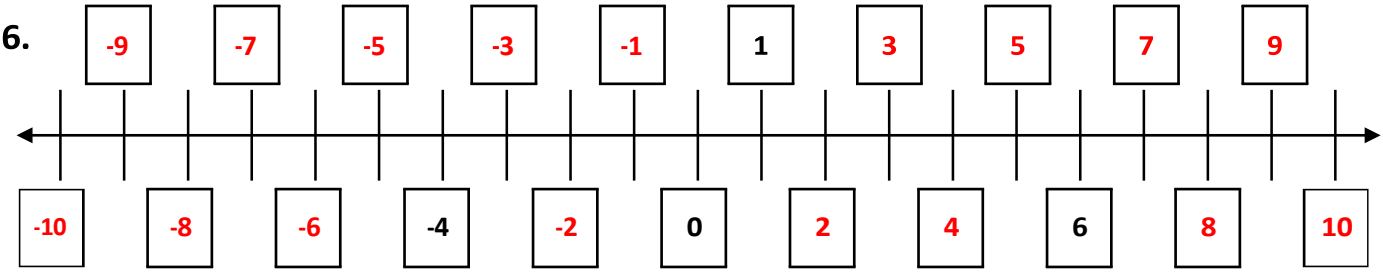
3. > > <

4. 900,009; 900,999; 909,990; 909,999; 990,009

5.

10	40	70
200	800	1,000
4,000	3,000	3,000
20,000	20,000	30,000

6.



7. –15˚C; -7˚C; -5˚C; 0˚C; 5˚C

8. 278,492

Number – Addition, Subtraction, Multiplication and Division

9. 83,208

10. 170,368

11. 256

12. 282.24

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13. 7 x 200 = 1,400 | 30 x 300 = 9,000

350 ÷ 50 = 7 | 3,600 ÷ 6 = 600

14. 1; 2; 3; 6

15. 36 and 72

16. 11; 13; 17; 19; 23

17. 36 + (8 x 9) = 108 | 6^2 + 3 + 9 x 2 = 57

(36 + 8) x 9 = 396 | 6^2 + (3 + 9) x 2 = 60

18. 144kg

19. 61 pence

20. 36,800 if numbers are rounded to the nearest 100.
37,000 if numbers are rounded to the nearest 1,000.

Fractions – (Decimals and Percentages)

21. $\frac{12}{36} = \frac{1}{3}$ $\frac{10}{90} = \frac{1}{9}$ $\frac{3}{12} = \frac{1}{4}$ $\frac{9}{45} = \frac{1}{5}$ $\frac{14}{56} = \frac{1}{4}$

22. $\frac{2}{5}$ $\frac{1}{2}$ $\frac{2}{3}$ $\frac{3}{4}$ $\frac{7}{8}$

23. $\frac{2}{3}$ > $\frac{7}{12}$ $\frac{2}{5}$ = $\frac{4}{10}$ $\frac{2}{3}$ < $\frac{7}{9}$

24. Accept equivalent fractions.

$\frac{7}{8}$ $\frac{5}{7}$

$\frac{2}{3}$ $\frac{13}{14}$

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25. $\frac{2}{3} \times \frac{1}{3} = \frac{2}{9}$ $\frac{1}{4} \times \frac{2}{5} = \frac{1}{10}$ $\frac{3}{7} \times \frac{1}{3} = \frac{1}{7}$

26. $\frac{3}{4} \div 4 = \frac{3}{16}$ $\frac{2}{5} \div 3 = \frac{2}{15}$ $\frac{3}{8} \div 5 = \frac{3}{40}$

27.

0.1	$\frac{1}{10}$
0.3	$\frac{3}{10}$
0.25	$\frac{1}{4}$
0.6	$\frac{6}{10}$
0.8	$\frac{4}{5}$

28.

	x 10	x 100	x 1,000
1.554	15.54	155.4	1,554
0.8	8	80	800
	÷ 10	÷ 100	÷ 1,000
65	6.5	0.65	0.065
137	13.7	1.37	0.137

29. 1.36 x 4 = 5.44 2.32 x 6 = 13.85 1.74 x 5 = 8.7

30. 17.4

31. 199.1

32. £138.25

33. 136m

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34. 45%

Ratio and Proportion

35. £3.65

36. £1.50

37. 15% of 200 = 30 5% of 80 = 4

30% of 75 = 22.5 60% of 350 = 210

38. £44.20

39. A = 6cm B = 12cm, Scale factor = 3

40. Taku gets £120; Arianna gets £100

Algebra

41. a = 25 n = 24 d = 10

42.

n	1	2	3	4	5	6	7	8	9
2n	2	4	6	8	10	12	14	16	18
3n + 1	4	7	10	13	16	19	22	25	28

43. 5 11 [17] 23 29 [35] 41 [47] 53 [59] Rule = + 6

[65] 50 35 [20] 5 [-10] -25 [-40] -55 [-70] Rule = – 15

2 [3] 5 9 17 [33] 65 [129] Rule = x 2 – 1

44. 8

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45. 44

46. a = 6 and b = 3 a

= 4 and b = 6 a

= 2 and b = 9

47. 19 + 18

Measurement

48. 60 cups

49. $1\frac{1}{2}$ l = 1,500 ml

1,350 ml = 1.35 l

$3\frac{1}{4}$ l = 3,250 ml

1,765 ml = 1.765 l

50. $2\frac{3}{4}$ kg = 2,750 g

12,420 g = 12.42 kg

$5\frac{1}{2}$ kg = 5,500 g

7,650 g = 7.65 kg

51.

mm	cm	m
500mm	50cm	0.5m
3,500mm	350cm	3.5m
250mm	25cm	0.25m
3,100mm	310cm	3.1m

52.

Approximately 1 mile = 1.6 kilometres	
3 miles	4.8 km
5 miles	8 km
6 miles	9.6 km
0.5 miles	0.8 km

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53. Same area = Shape A and E

    Same perimeter = Shape C and F

54. 17cm

55. 9m

56. $48cm^2$

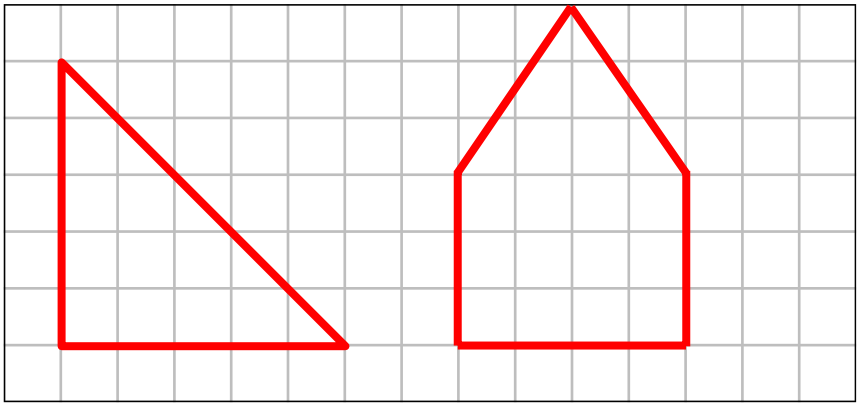
57. $24.5m^2$

58. A = $14cm^3$ B = $16cm^3$ C = $17cm^3$

    Shape C has the largest volume.

Properties of Shape

59 and 60.



61.

3D shape	Faces	Vertices	Edges
Cylinder	2	0	2
Sphere	0	0	0
Tetrahedron	4	4	6
Square Based pyramid	5	5	8

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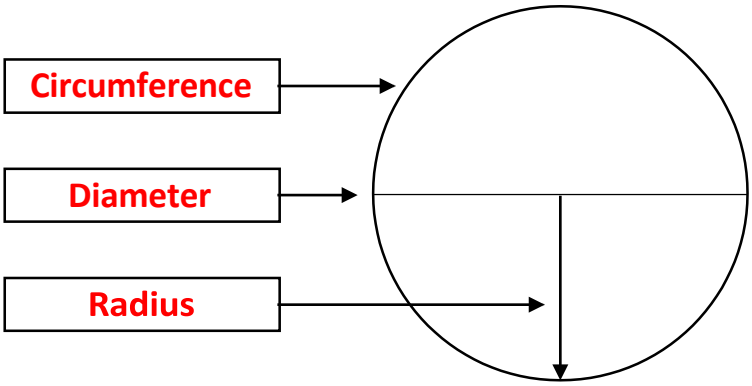
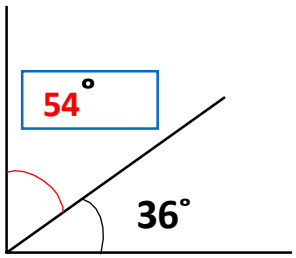
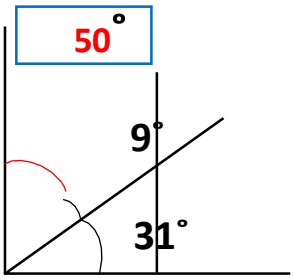
62. A = 50º B = 96º

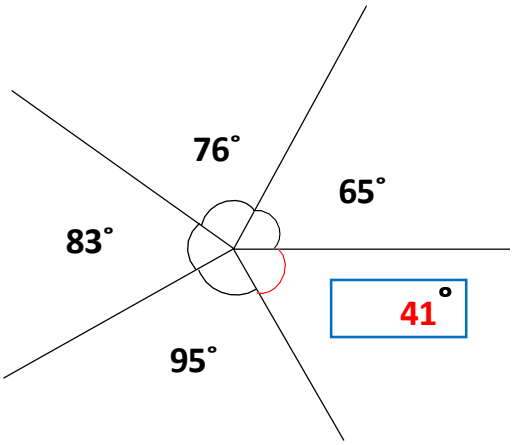
63.

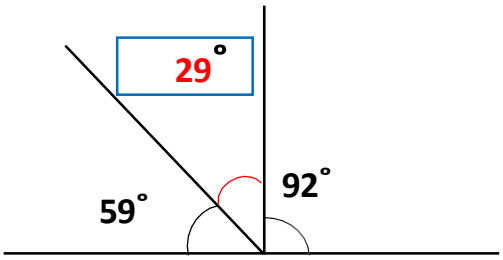


64.









Geometry – Position and Direction

65. A = (2,4); B = (5,4); C = (2,1); D = (5,1)

66. A = (-2,4); B = (-5,4); C = (-2,1); D = (-5,1)

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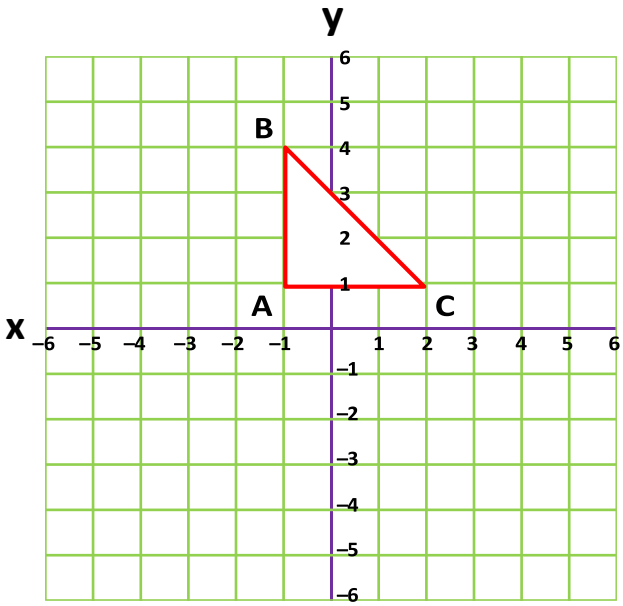
67.



68. **A = (-1,1) B = (-1,4) C = (2,1)**

Statistics

69. **12; 6; 48; $\frac{1}{8}$**

70. **15km; Blue car; 20km; 4 hours**

71. **31**